

Classical and Quantum Sensing

Noise, Measurements, and Fundamental Limits

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Motivation

Sensing aims to estimate physical parameters such as:

- ▶ Optical phase, frequency, and intensity
- ▶ Magnetic and gravitational fields
- ▶ Forces and displacements

Key question: What fundamentally limits how precisely these quantities can be measured?

Classical Sensing

In classical sensing:

- ▶ Signals are treated as continuous variables
- ▶ Noise is modeled statistically
- ▶ Measurement back-action is neglected

Examples include:

- ▶ Optical power meters
- ▶ Electrical voltage and current sensors
- ▶ Classical interferometers

Photodetection: Classical Picture

Classically, light is an electromagnetic wave:

$$\mathbf{E}(t) = E_0 \cos(\omega t + \phi)$$

Photodetectors are modeled as:

- ▶ Square-law devices
- ▶ Output proportional to intensity $I \propto |E|^2$

This approximation fails at low photon numbers.

Classical Noise

Noise represents unavoidable random fluctuations.

Key characteristics:

- ▶ Limits sensitivity and resolution
- ▶ Often described by power spectral density
- ▶ Can be intrinsic or technical

Johnson (Thermal) Noise

Johnson noise arises from thermal motion of charge carriers.

Voltage noise spectral density:

$$S_V(f) = 4k_B TR$$

- ▶ Independent of frequency (white noise)
- ▶ Exists even without applied voltage
- ▶ Fundamental for resistive elements

Telegraph Noise

Telegraph noise consists of random switching between discrete levels.

Physical origin:

- ▶ Charge trapping and detrapping
- ▶ Two-level fluctuators

Produces excess low-frequency noise and non-Gaussian statistics.

Flicker (1/f) Noise

Flicker noise has power spectral density:

$$S(f) \propto \frac{1}{f^\alpha}, \quad \alpha \approx 1$$

- ▶ Dominates at low frequencies
- ▶ Ubiquitous in electronic and optical systems
- ▶ Limits long-term stability

Sensitivity of Classical Measurements

Sensitivity quantifies the smallest resolvable parameter change.

Depends on:

- ▶ Noise amplitude
- ▶ Measurement bandwidth
- ▶ Integration time

Longer averaging improves precision only until low-frequency noise dominates.

Classical Fisher Information

Given outcomes x with probability $p(x|\theta)$:

$$F(\theta) = \int dx p(x|\theta) \left(\frac{\partial}{\partial \theta} \ln p(x|\theta) \right)^2$$

Interpretation:

- ▶ Measures how sensitive outcomes are to parameter changes
- ▶ Larger Fisher information implies higher achievable precision

Cramér–Rao Bound

For any unbiased estimator $\hat{\theta}$:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{F(\theta)}$$

Consequences:

- ▶ Fundamental bound independent of estimator choice
- ▶ Saturated only by optimal measurements and estimators

Quantum Measurements

In quantum mechanics:

- ▶ Measurement outcomes are probabilistic
- ▶ Measurement changes the system state
- ▶ Observer and system cannot be separated

This leads to fundamentally new limits on sensing.

Projective Measurements

Projective measurements correspond to Hermitian operators.

Measurement operators:

$$\Pi_i = |i\rangle\langle i|$$

- ▶ Outcomes correspond to eigenvalues
- ▶ State collapses into an eigenstate
- ▶ Maximal disturbance

Generalized (POVM) Measurements

Positive Operator-Valued Measures (POVMs):

$$\sum_i E_i = \mathbb{I}$$

Allow:

- ▶ Non-orthogonal outcomes
- ▶ Joint measurements of incompatible observables
- ▶ Optimal quantum estimation

Weak and Continuous Measurements

Weak measurements extract partial information.

Key features:

- ▶ Small back-action per measurement
- ▶ Information accumulates over time
- ▶ Described by stochastic master equations

Error–Disturbance Relations

Measurement precision is fundamentally linked to disturbance.

Generalized uncertainty relations:

- ▶ Extend Heisenberg uncertainty principle
- ▶ Govern trade-offs in quantum metrology

Standard Quantum Limit (SQL)

SQL arises from:

- ▶ Measurement imprecision noise
- ▶ Quantum back-action noise

Precision scaling:

$$\Delta\theta \propto \frac{1}{\sqrt{N}}$$

Typical of coherent states and classical strategies.

Quantum Non-Demolition Measurements

QND measurements:

- ▶ Measure observables commuting with the Hamiltonian
- ▶ Avoid back-action on measured quantity
- ▶ Enable repeated measurements

Critical for gravitational-wave detectors and atomic clocks.

Fock States

Fock states have definite photon number:

$$|n\rangle$$

Properties:

- ▶ Zero photon-number uncertainty
- ▶ Completely undefined phase
- ▶ Strongly non-classical

Coherent States

Coherent states:

$$|\alpha\rangle$$

Properties:

- ▶ Poissonian photon statistics
- ▶ Minimum uncertainty states
- ▶ Behave most classically

Squeezed States

Squeezed states redistribute quantum noise.

- ▶ Noise reduced in one quadrature
- ▶ Increased noise in conjugate quadrature
- ▶ Enable sub-SQL sensitivity

Quantum State Tomography

Quantum tomography reconstructs the density matrix.

Requires:

- ▶ Many repeated measurements
- ▶ Different measurement bases
- ▶ Statistical reconstruction algorithms

Phase-Space Distributions

Phase-space representations:

- ▶ Wigner function (can be negative)
- ▶ Glauber–Sudarshan P -function
- ▶ Husimi Q -function

Used to visualize non-classicality.

Quantum Photodetection

Photodetection is inherently quantum.

Key features:

- ▶ Discrete detection events
- ▶ Shot noise is fundamental
- ▶ Governed by quantum measurement theory

Square-Law and Intensity Detection

Intensity detectors measure photon number.

- ▶ Sensitive to $\langle \hat{n} \rangle$
- ▶ Cannot access phase directly
- ▶ Limited by photon shot noise

Linear Detectors and Quadrature Measurements

Homodyne detection:

- ▶ Measures field quadratures
- ▶ Phase-sensitive detection
- ▶ Essential for squeezed states

Single-Photon Sensing

Applications:

- ▶ Low-light imaging
- ▶ Quantum lidar
- ▶ High-contrast measurements

Advantages:

- ▶ Reduced background noise
- ▶ High information per photon

Entanglement-Based Sensing

Entanglement enables:

$$\Delta\theta \propto \frac{1}{N}$$

Used in:

- ▶ Quantum interferometry
- ▶ Precision clocks
- ▶ Phase estimation

Atomic State-Based Sensing

Atomic sensors exploit:

- ▶ Long coherence times
- ▶ Well-controlled quantum states

Applications:

- ▶ Atomic clocks
- ▶ Atom interferometers

Solid-State Spin-Based Sensing

Solid-state spins (e.g., NV centers):

- ▶ Operate at room temperature
- ▶ Enable nanoscale resolution

Applications:

- ▶ Magnetometry
- ▶ Gravimetry

Summary

- ▶ Classical sensing limited by noise and statistics
- ▶ Quantum measurements introduce back-action
- ▶ Non-classical states surpass classical limits
- ▶ Quantum sensing enables unprecedented precision